Error partitioning: synthetic

400 chips, 400 with a verdict; predicted classes ['bar', 'cross', 'disc', 'ring']; verdict classes ['bar', 'cross', 'disc', 'ignore', 'ring', 'triangle']; new to the model: ['triangle']

Pictures from the dinov2 embedding; the first table compares every embedding run. 2026-10-14 15:05. Percentages throughout; +/- is one standard deviation over random draws. Under every chip: its verdict, and a bar in the verdict's colour.

bar cross disc ignore ring triangle

The embeddings compared

embedding	all chips, k=10 k-means purity	3 seeds/class, spreading	cluster then label (k=10): clicks / accuracy	new-class AUROC
pixels	53	48 +/- 3	10 / 47	35 +/- 6
resnet18	98	94 +/- 6	10 / 99	88 +/- 3
dinov2	100	95 +/- 4	10 / 100	95 +/- 2

Clustering by predicted class, over-clustered (dinov2)

Purity: the accuracy if each cluster took its majority verdict. Clicks: the verdicts that costs. An HDBSCAN chip left unclustered is a click of its own.

predicted	chips	kinds	k	k-means purity	clicks	spectral purity	clicks	HDBSCAN purity	clicks
bar	89	4	10	99	10	100	10	100	8 (3 alone)
cross	129	6	10	100	10	100	10	100	19 (10 alone)
disc	88	5	10	100	10	99	10	100	8 (2 alone)
ring	94	4	10	100	10	100	10	73	3 (0 alone)
all	400	6	10	100	10	100	10	100	23 (14 alone)

Seeded labelling (dinov2)

A few verdicts per class, the rest filled in. Accuracy on the chips not seeded; coverage is the share of them that can be taken at 90 % precision by trusting only the confident ones.

seeds per class	label spreading	kNN vote	logistic head	coverage at 90 % (spreading / logistic)
1	79 +/- 13	22 +/- 0	89 +/- 4	83 / 96
2	87 +/- 10	60 +/- 3	94 +/- 4	94 / 98
3	95 +/- 4	97 +/- 2	97 +/- 2	100 / 100
5	100 +/- 0	98 +/- 1	99 +/- 1	100 / 100
10	100 +/- 0	99 +/- 1	99 +/- 1	100 / 100

Cluster, then label (dinov2)

The reviewer names the most central chip of each cluster and the names spread to the rest.

clustering	clicks	accuracy	coverage at 90 %
kmeans	5	81	89
spectral	5	90	100
kmeans (k=10)	10	100	100
spectral (k=10)	10	100	100
hdbscan	9	100	100

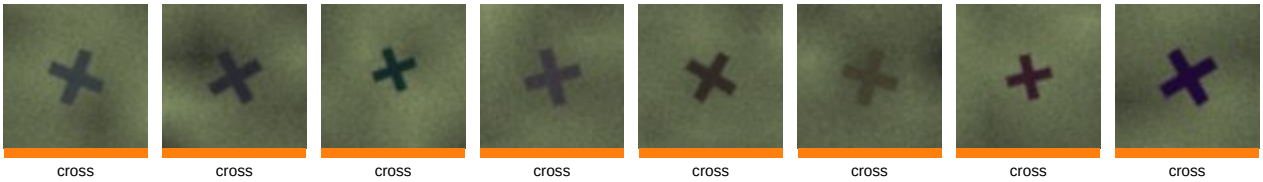
Novelty (dinov2)

AUROC for telling a chip of a class the model does not know (40 chips) from one of a known class (320), by distance to the nearest known prototype 95 +/- 2 and to the nearest known seeds 96 +/- 2. 50 is chance, 100 perfect.

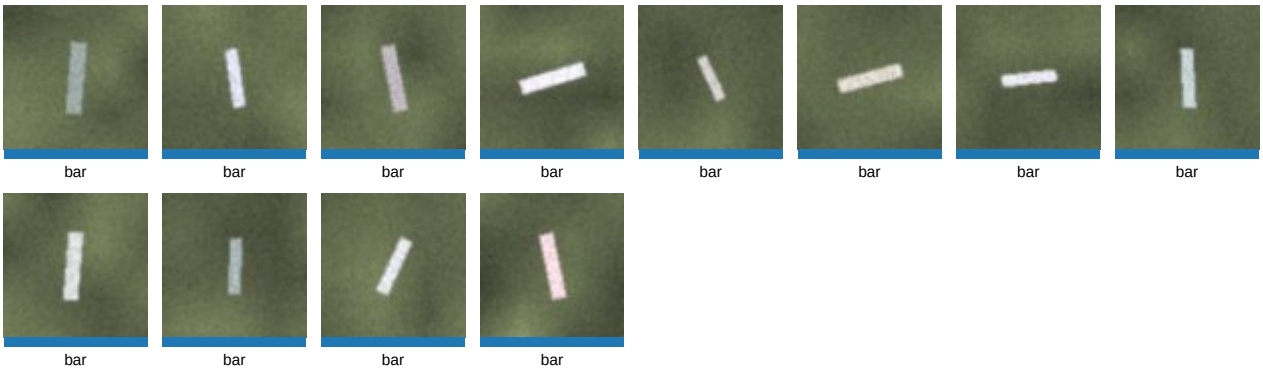
Predicted bar: 89 chips in 10 clusters (k-means)

Verdicts: bar 31, cross 35, ignore 9, triangle 14. Purity 99 for 10 clicks. Each cluster's chips in order of distance from its centre; a red frame marks a chip whose verdict is not the cluster's majority.

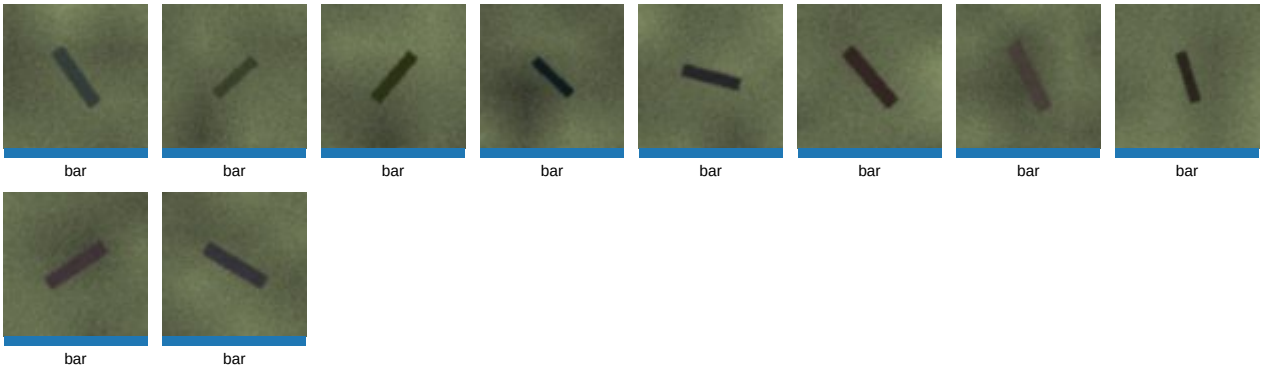
Cluster 4: 13 chips, majority cross (13 of 13)



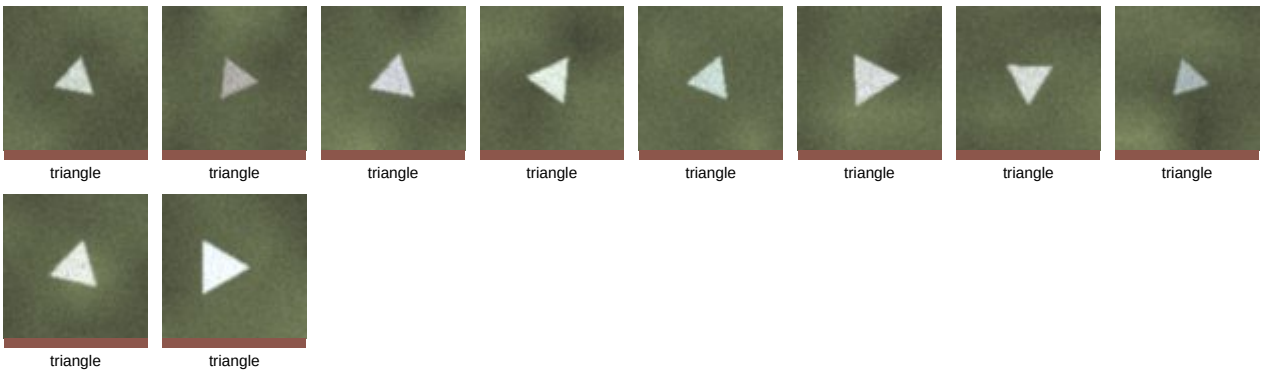
Cluster 9: 12 chips, majority bar (12 of 12)



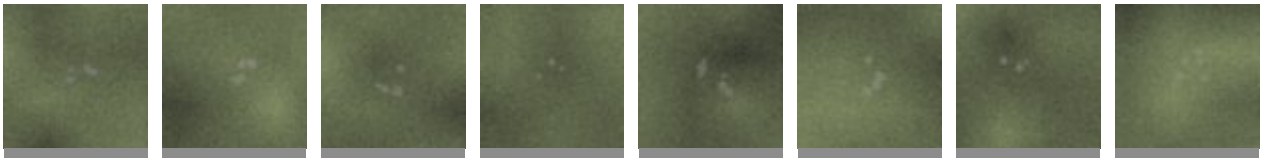
Cluster 10: 10 chips, majority bar (10 of 10)



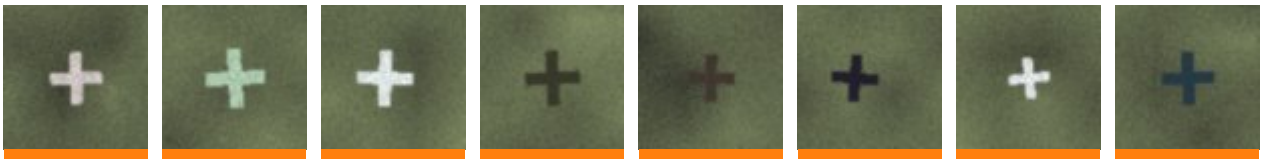
Cluster 7: 10 chips, majority triangle (10 of 10)



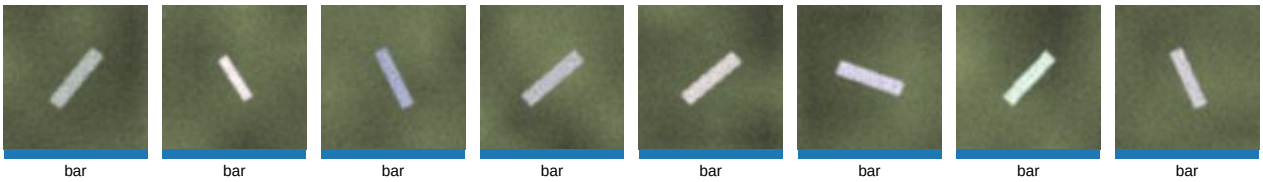
Cluster 6: 10 chips, majority ignore (9 of 10)



Cluster 8: 8 chips, majority cross (8 of 8)



Cluster 3: 8 chips, majority bar (8 of 8)



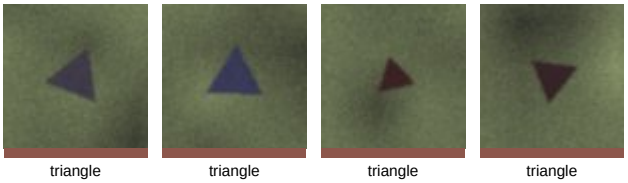
Cluster 5: 7 chips, majority cross (7 of 7)



Cluster 2: 7 chips, majority cross (7 of 7)



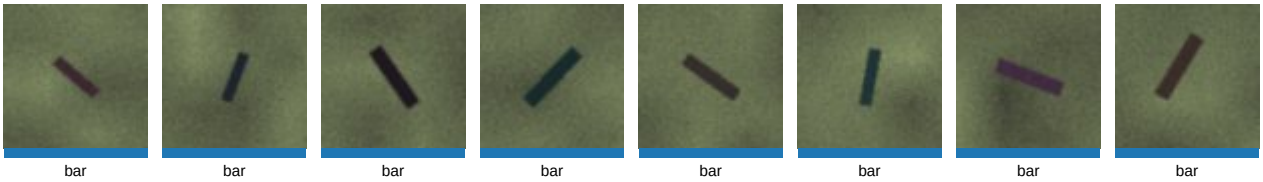
Cluster 1: 4 chips, majority triangle (4 of 4)



Predicted cross: 129 chips in 10 clusters (k-means)

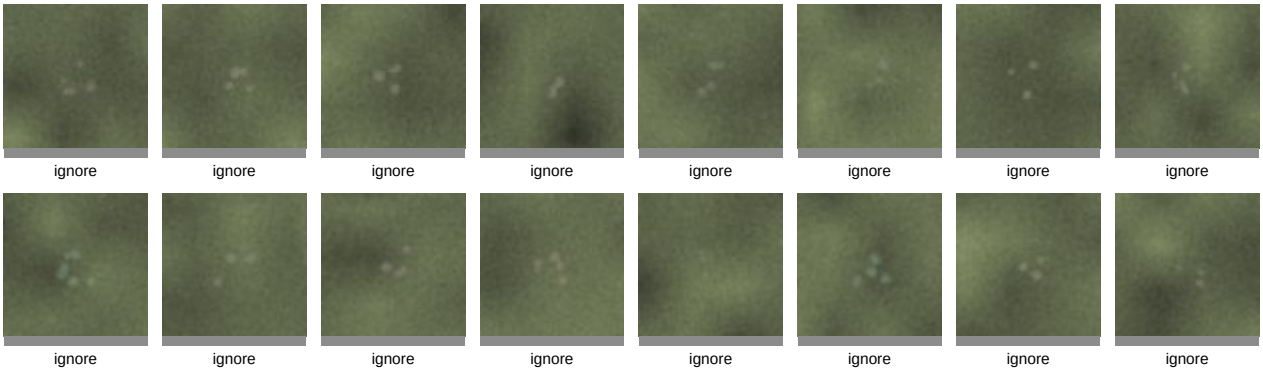
Verdicts: bar 47, cross 23, disc 10, ignore 16, ring 14, triangle 19. Purity 100 for 10 clicks. Each cluster's chips in order of distance from its centre; a red frame marks a chip whose verdict is not the cluster's majority.

Cluster 10: 19 chips, majority bar (19 of 19)

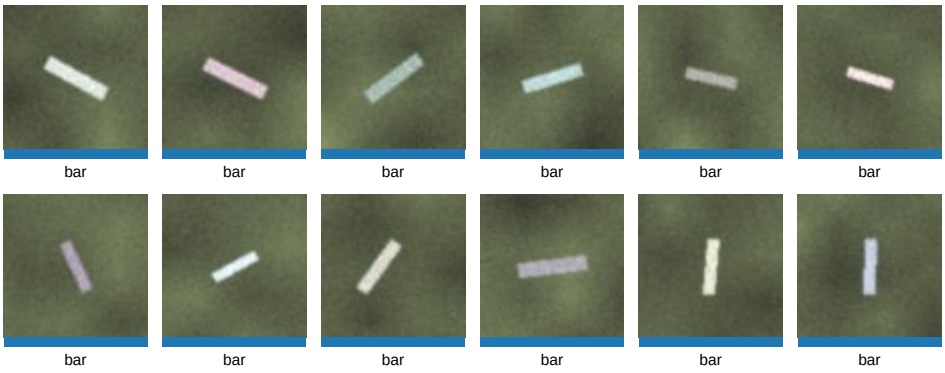


... and 3 more

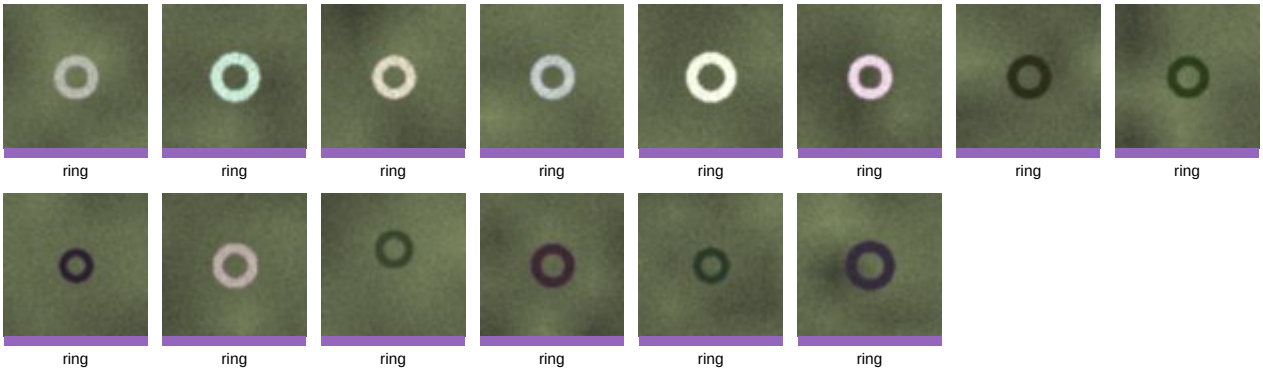
Cluster 4: 16 chips, majority ignore (16 of 16)



Cluster 8: 14 chips, majority bar (14 of 14)



Cluster 2: 14 chips, majority ring (14 of 14)



Cluster 1: 14 chips, majority bar (14 of 14)



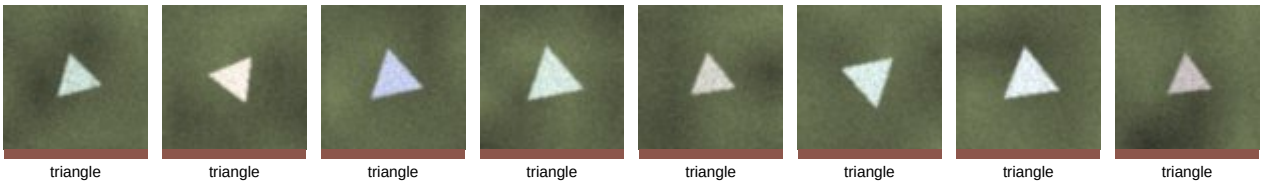


Cluster 6: 12 chips, majority cross (12 of 12)





Cluster 5: 11 chips, majority triangle (11 of 11)



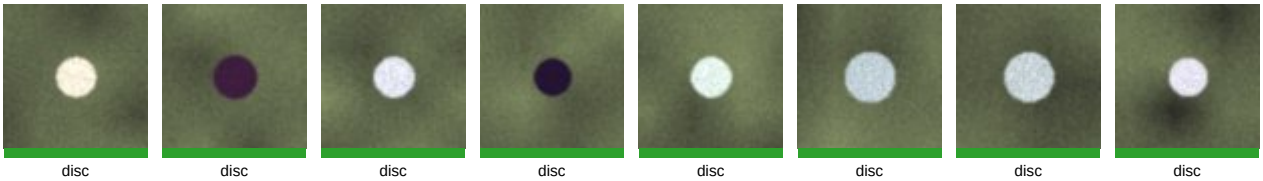


Cluster 3: 11 chips, majority cross (11 of 11)





Cluster 9: 10 chips, majority disc (10 of 10)





disc disc

Cluster 7: 8 chips, majority triangle (8 of 8)

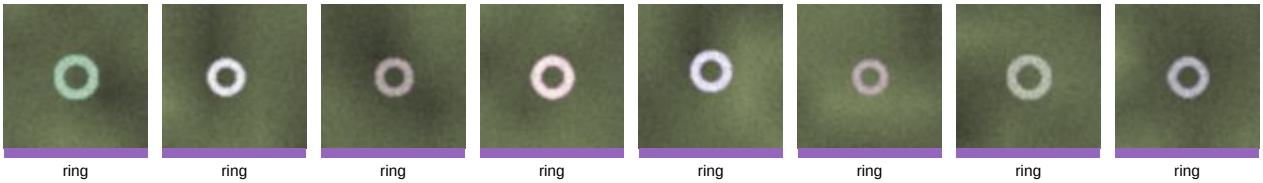


triangle triangle triangle triangle triangle triangle triangle triangle

Predicted disc: 88 chips in 10 clusters (k-means)

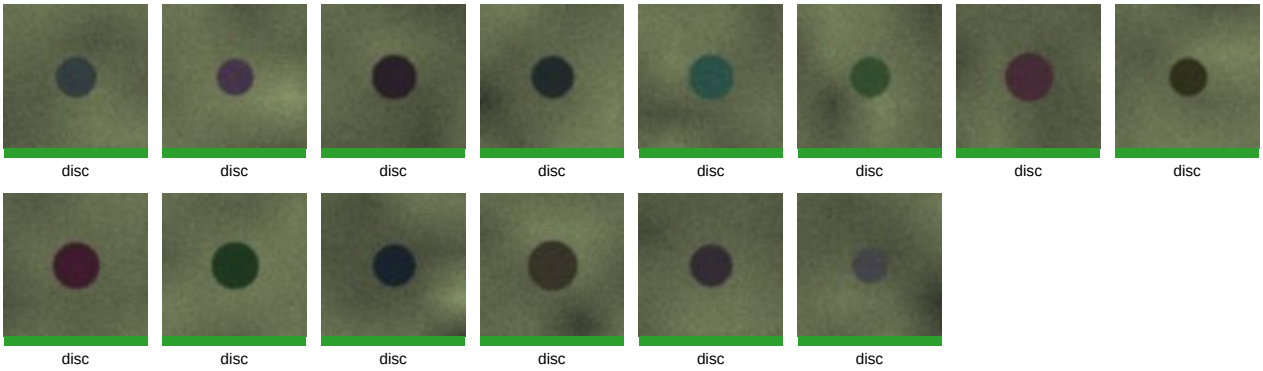
Verdicts: bar 10, disc 33, ignore 5, ring 33, triangle 7. Purity 100 for 10 clicks. Each cluster's chips in order of distance from its centre; a red frame marks a chip whose verdict is not the cluster's majority.

Cluster 1: 20 chips, majority ring (20 of 20)

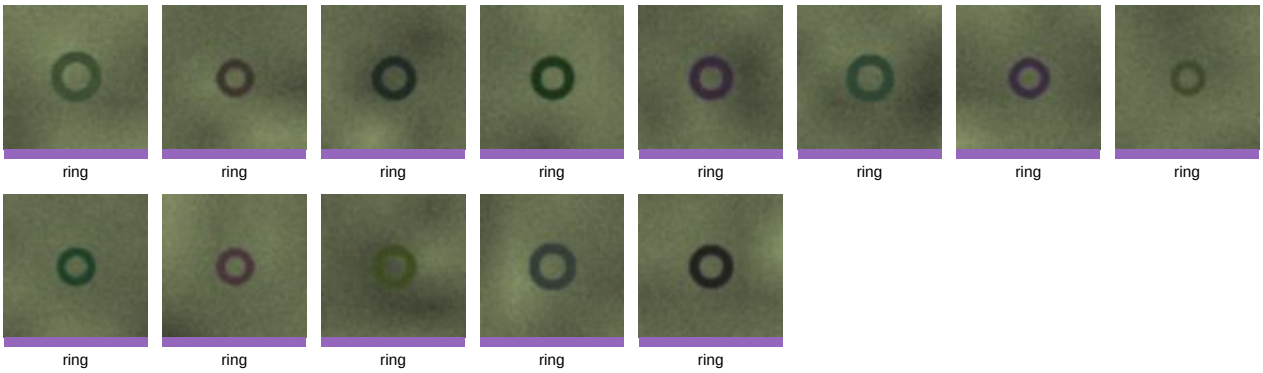


... and 4 more

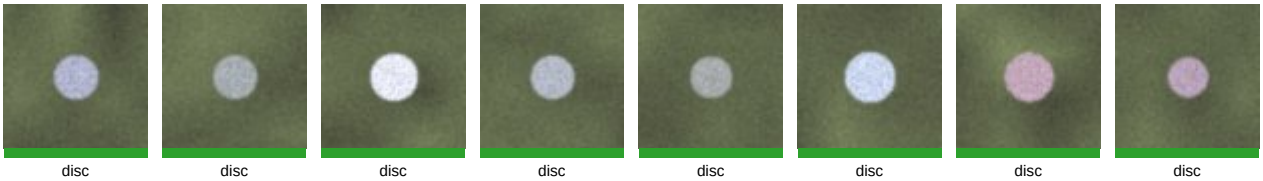
Cluster 7: 14 chips, majority disc (14 of 14)



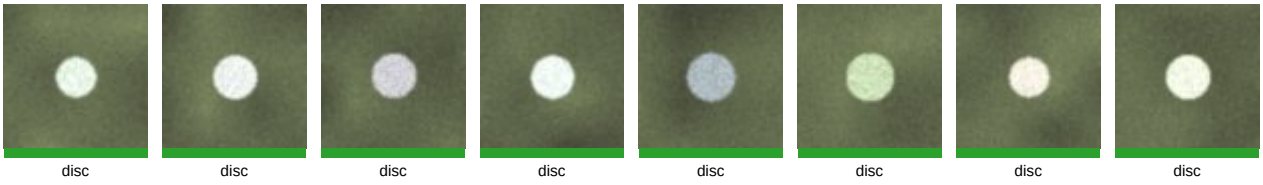
Cluster 5: 13 chips, majority ring (13 of 13)



Cluster 9: 8 chips, majority disc (8 of 8)



Cluster 8: 8 chips, majority disc (8 of 8)



Cluster 4: 7 chips, majority triangle (7 of 7)



triangle triangle triangle triangle triangle triangle triangle

Cluster 10: 5 chips, majority ignore (5 of 5)



ignore ignore ignore ignore ignore

Cluster 6: 5 chips, majority bar (5 of 5)



bar bar bar bar bar

Cluster 2: 5 chips, majority bar (5 of 5)



bar bar bar bar bar

Cluster 3: 3 chips, majority disc (3 of 3)

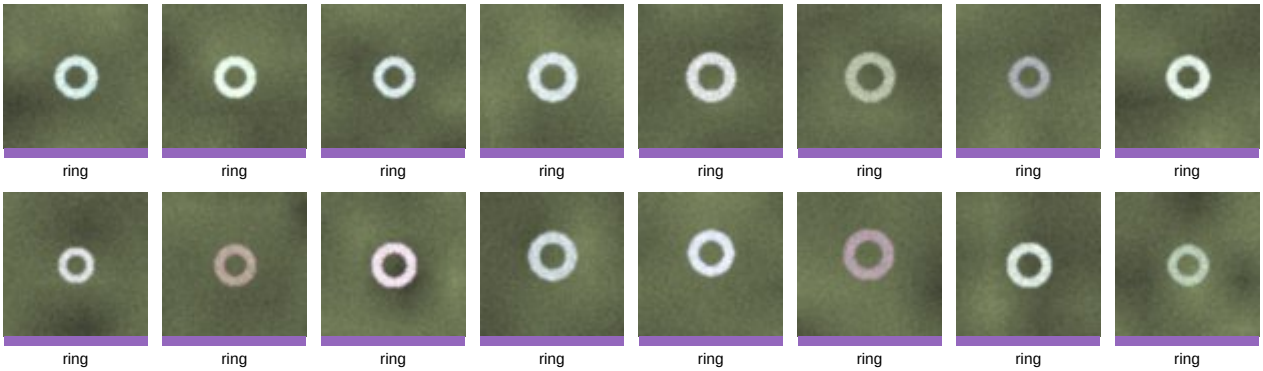


disc disc disc

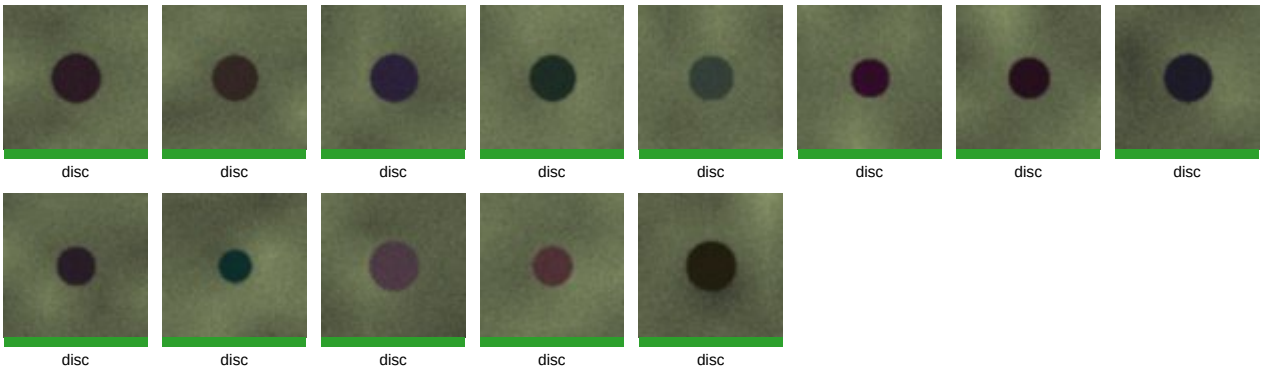
Predicted ring: 94 chips in 10 clusters (k-means)

Verdicts: cross 14, disc 45, ignore 10, ring 25. Purity 100 for 10 clicks. Each cluster's chips in order of distance from its centre; a red frame marks a chip whose verdict is not the cluster's majority.

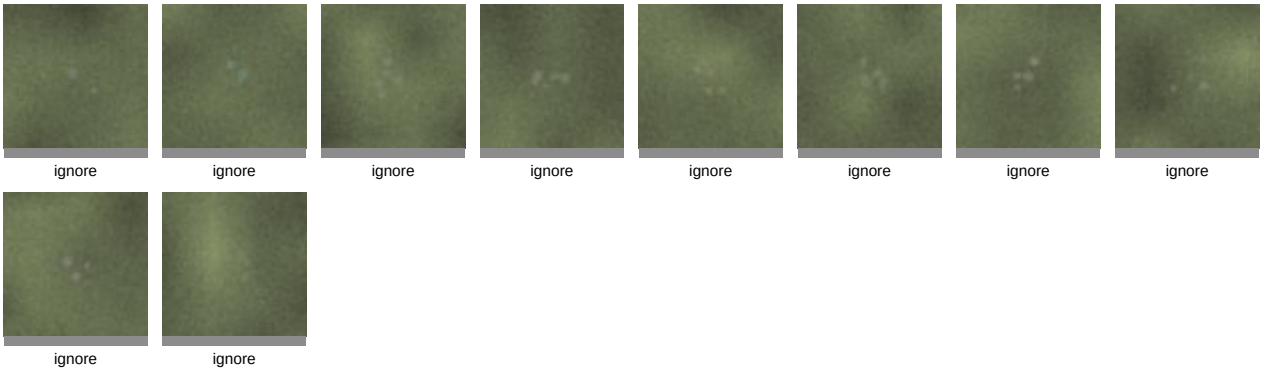
Cluster 4: 16 chips, majority ring (16 of 16)



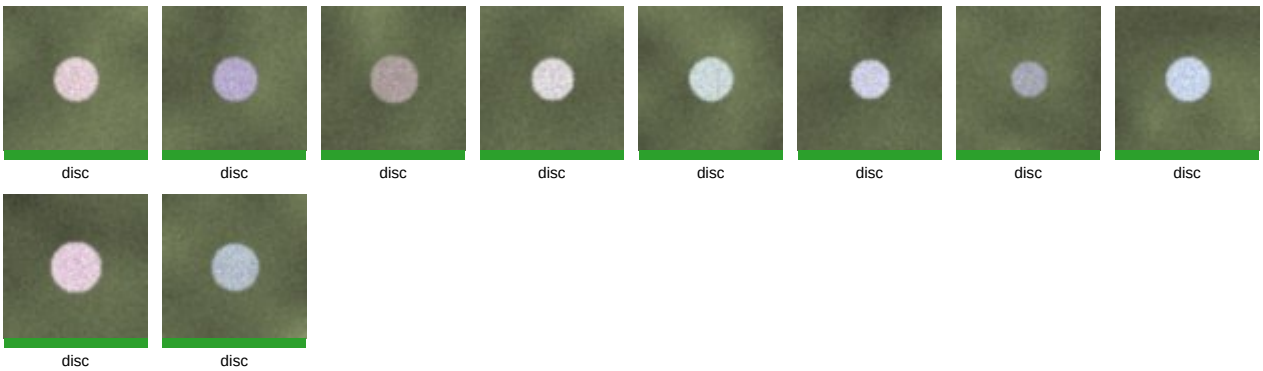
Cluster 6: 13 chips, majority disc (13 of 13)



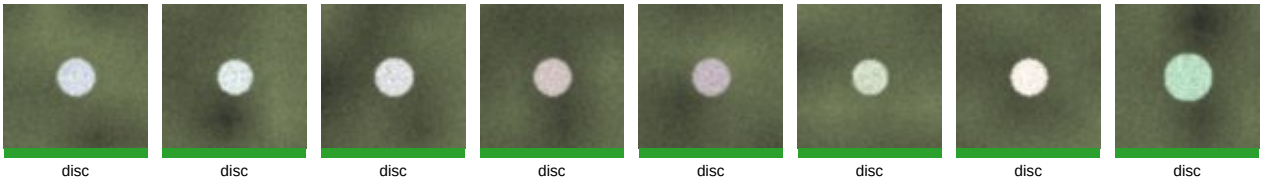
Cluster 5: 10 chips, majority ignore (10 of 10)



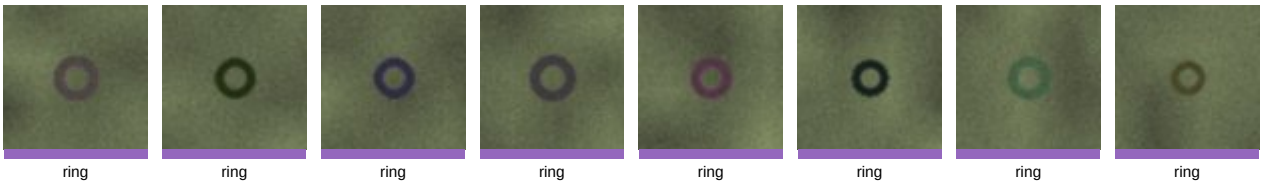
Cluster 3: 10 chips, majority disc (10 of 10)



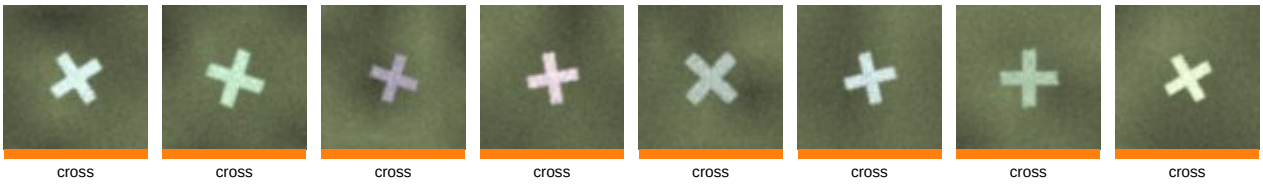
Cluster 1: 10 chips, majority disc (10 of 10)



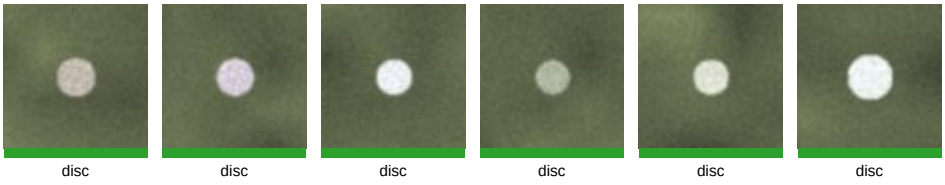
Cluster 7: 9 chips, majority ring (9 of 9)



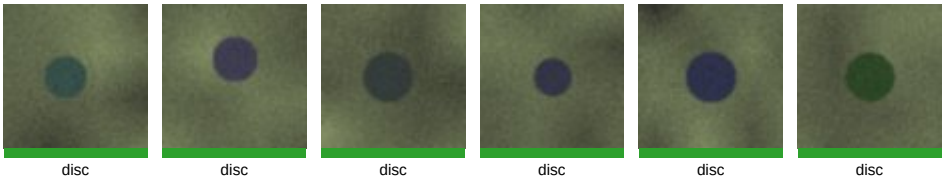
Cluster 2: 9 chips, majority cross (9 of 9)



Cluster 10: 6 chips, majority disc (6 of 6)



Cluster 9: 6 chips, majority disc (6 of 6)



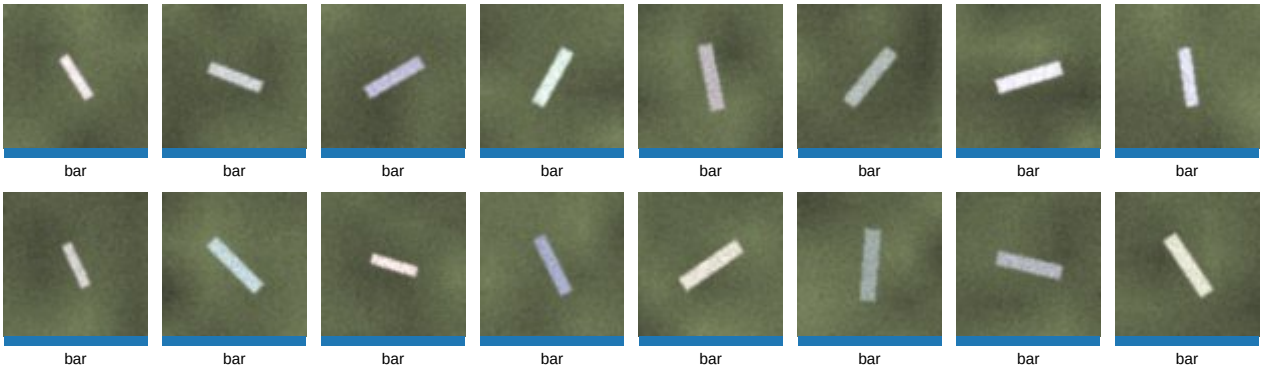
Cluster 8: 5 chips, majority cross (5 of 5)



All chips: 400 chips in 10 clusters (k-means)

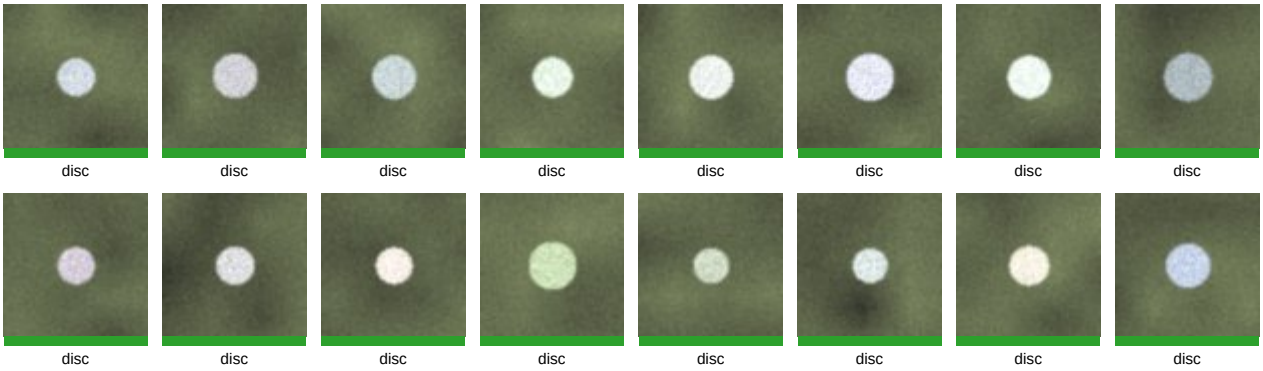
Verdicts: bar 88, cross 72, disc 88, ignore 40, ring 72, triangle 40. Purity 100 for 10 clicks. Each cluster's chips in order of distance from its centre; a red frame marks a chip whose verdict is not the cluster's majority.

Cluster 8: 54 chips, majority bar (54 of 54)



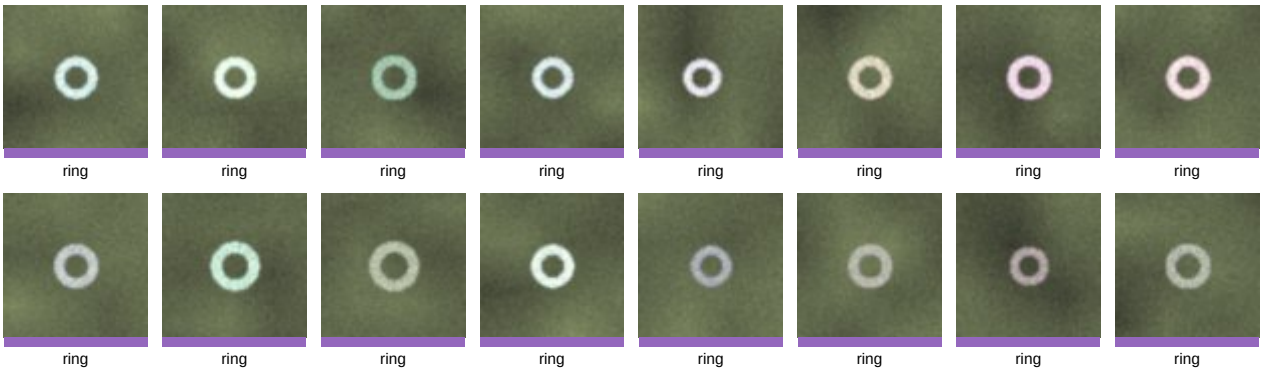
... and 38 more

Cluster 4: 51 chips, majority disc (51 of 51)



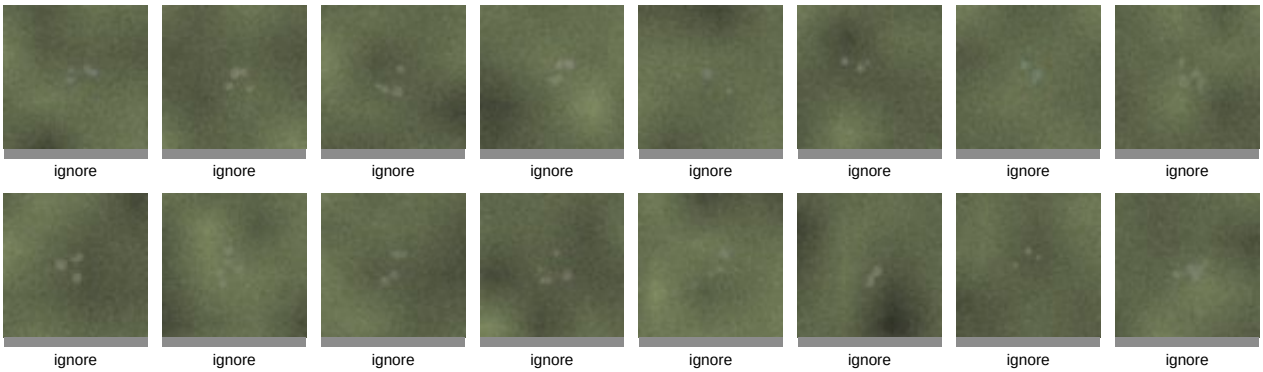
... and 35 more

Cluster 1: 43 chips, majority ring (43 of 43)



... and 27 more

Cluster 6: 40 chips, majority ignore (40 of 40)



... and 24 more

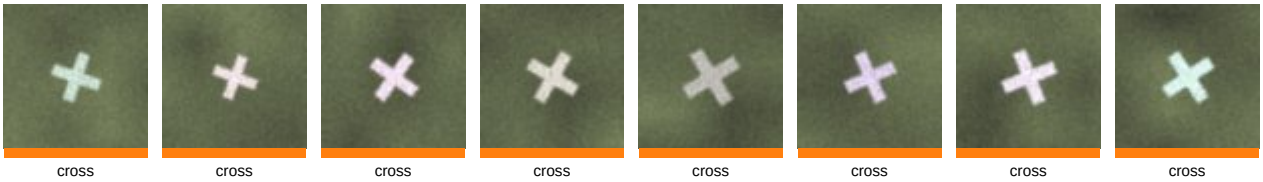
Cluster 5: 40 chips, majority triangle (40 of 40)

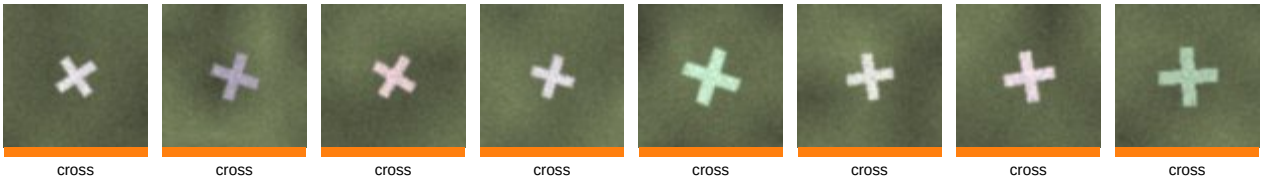




... and 24 more

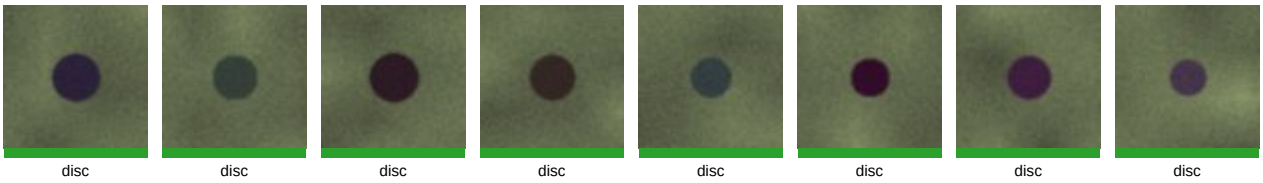
Cluster 9: 39 chips, majority cross (39 of 39)

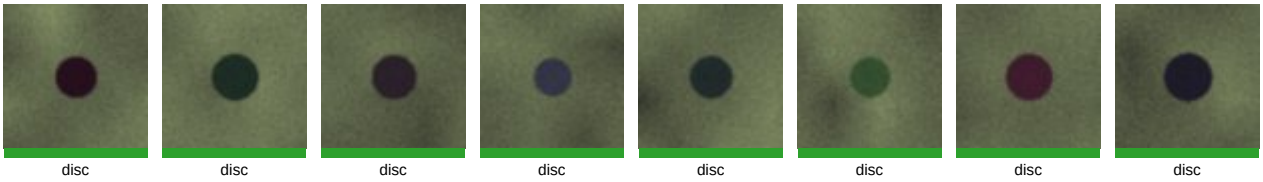




... and 23 more

Cluster 7: 37 chips, majority disc (37 of 37)

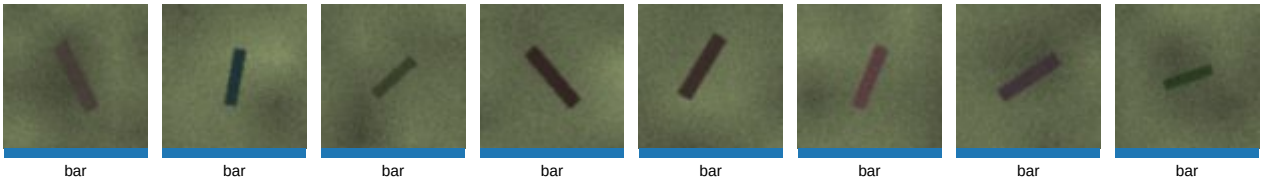




... and 21 more

Cluster 2: 34 chips, majority bar (34 of 34)





... and 18 more

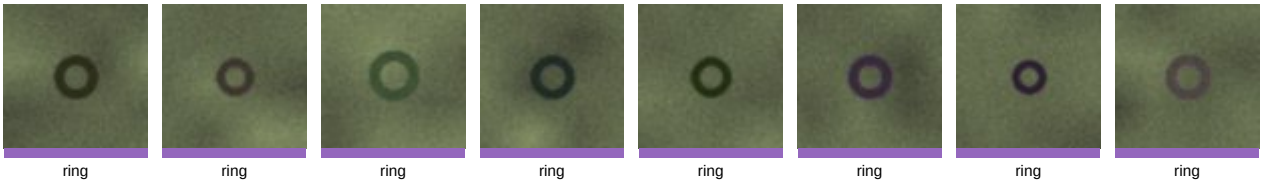
Cluster 3: 33 chips, majority cross (33 of 33)

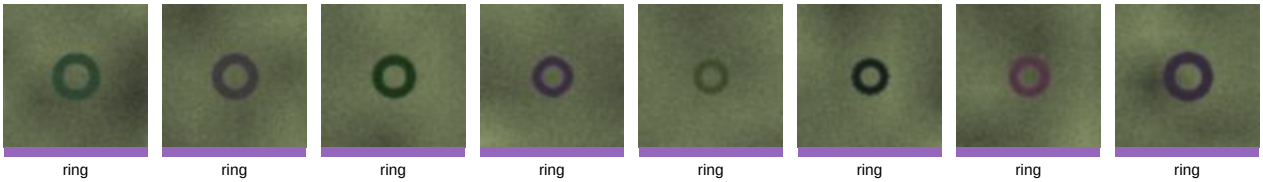




... and 17 more

Cluster 10: 29 chips, majority ring (29 of 29)



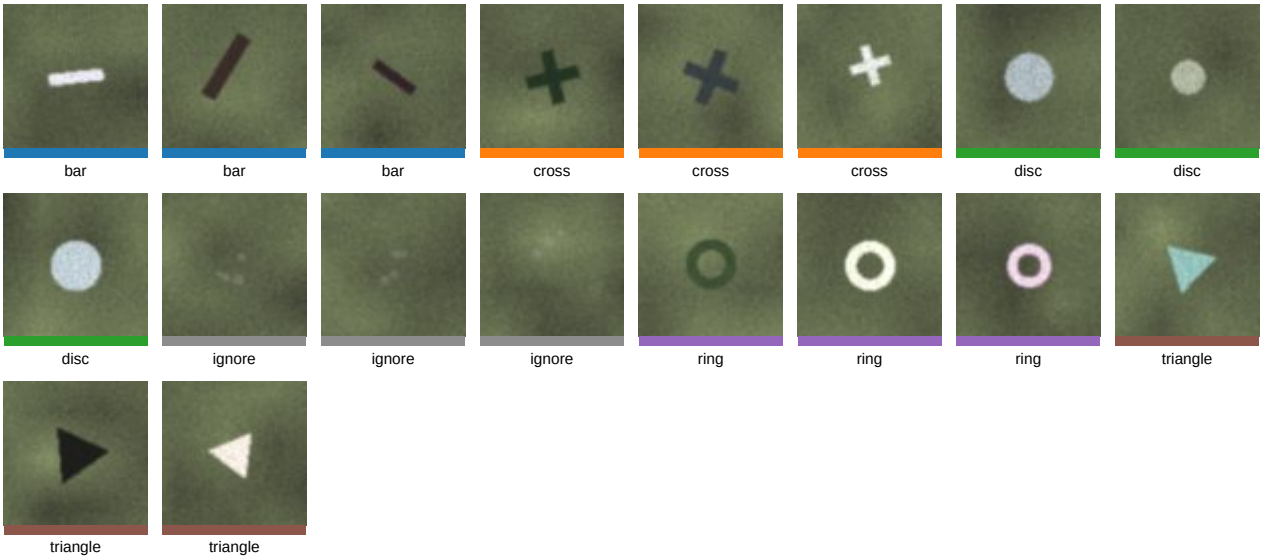


... and 13 more

3 verdicts per class, the rest spread

18 seeds; 382 chips filled in by label spreading, 381 right (100), 1 wrong. One random draw; the table on the cover averages ten.

The seeds



Filled in wrongly (1): captioned with what spreading called it; the bar is the verdict's colour



Farthest from the known classes

The chips least like any known-class prototype (five seeds per known class), farthest first: what a reviewer should see first when a new class may be hiding in a review. An orange frame marks a chip whose verdict is a class the model does not predict; a grey bar, Ignore.

